Waterbirds IMRaD Report

Section 1: Introduction

Standard vision models trained via Empirical Risk Minimization (ERM) optimize overall average loss over the training distribution. When non-causal background cues correlate strongly with object classes, gradient descent prioritizes opportunistic background features rather than robust semantic representations. In dataset environments with severe subgroup imbalance, ERM minimizes overall loss while collapsing on under-represented sub-populations.

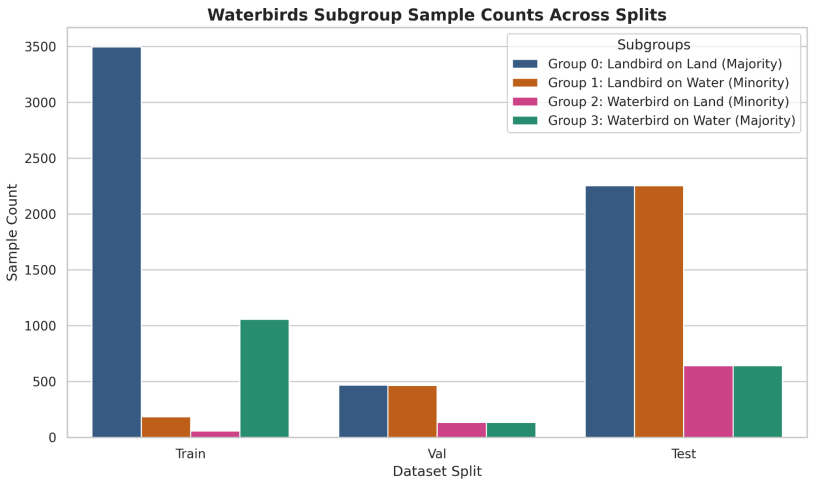
To evaluate and mitigate spurious background correlation, the Waterbirds benchmark partitions the classification problem into four mutually exclusive subgroups defined by bird class ($y \in$ {Landbird, Waterbird}) and background scene ($a \in$ {Land, Water}):

- Group 0 (y=0, a=0): **Landbird on Land** (Majority / Spurious, N_{train} = 3,498)
- Group 1 (y=0, a=1): **Landbird on Water** (Minority / Out-of-Distribution, N_{train} = 184)
- Group 2 (y=1, a=0): **Waterbird on Land** (Minority / Out-of-Distribution, N_{train} = 56)
- Group 3 (y=1, a=1): **Waterbird on Water** (Majority / Spurious, N_{train} = 1,057)

Because overall average accuracy masks subpopulation failure, the primary evaluation metric across all experiments is Worst-Group Accuracy ($Acc_{worst} = \min_g Acc_g$), supplemented by Average Accuracy (Acc_{avg}) to assess global performance trade-offs.

Section 2: Methods and Materials

We utilize the Waterbirds benchmark dataset (Sagawa et al., 2020), which crops bird foregrounds from the CUB-200-2011 dataset and pastes them onto land and water background images extracted from Places365. The dataset consists of 4,795 training images, 1,199 validation images, and 5,794 test images.



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To analyze backbone resilience against shortcut learning, we implement two pre-trained architectures:

1. **ResNet-50**: A 50-layer deep convolutional network with residual connections, modified by replacing the final classification head with a 2-class output layer.
2. **ViT-Base (Vision Transformer)**: A vision transformer using 16x16 patch embeddings and global multi-head self-attention mechanisms.

We evaluate three optimization objectives:

1. **Empirical Risk Minimization (ERM)**: Standard cross-entropy loss over all mini-batch samples without group awareness.
2. **Group-Balanced Loss Re-weighting**: Scales sample losses by inverse group frequencies ($w_g \propto 1 / N_g$), yielding a weight multiplier of 21.41x for Group 2 relative to 0.34x for Group 0.
3. **Group DRO (Distributionally Robust Optimization)**: Dynamically updates a group weight vector q via exponentiated gradient updates (step size η) to minimize worst-group loss: $L_{DRO}(\theta) = \sum_{\{g \in G\}} q_g * l_g(\theta)$

To isolate the effect of batch composition, models are trained under two data loading schemes:

- **Unbalanced Loader**: Standard random sampling (shuffle=True) reflecting training set proportions.
- **Balanced Loader**: WeightedRandomSampler enforcing equal subgroup representation per mini-batch.

All training is conducted on the NVIDIA Tesla P100 GPU from the Gorina4 server using SGD (learning rate 1e-4, momentum 0.9, weight decay 1e-4, batch size 32). Validation worst-group accuracy is monitored after each epoch for model checkpoint selection.

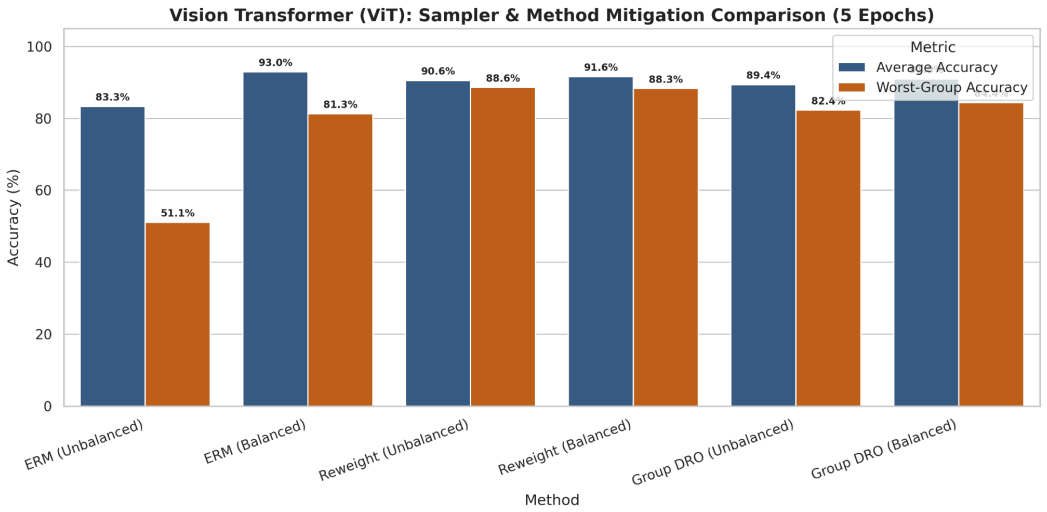
Section 3: Results

The table details test performance across backbones, mitigation objectives, and data sampling strategies. Except for these, all models were trained on the same settings (5 epochs, learning rate 1e-4 and weight decay 1e-4. Standard ERM on ResNet-50 collapses on Group 2 (3.43% Worst-Group Acc), whereas ViT-Base combined with Group DRO and balanced sampling achieves the top overall performance (86.29% Worst-Group Acc, 91.61% Avg Acc).

Architecture	Method	Sampler	Worst Acc	Avg Acc	G0	G1	G2	G3
ResNet-50	ERM Baseline	Unbalanced	8.88%	73.33%	99.69%	62.93%	8.88%	81.78%
ResNet-50	Loss Re-weighting	Unbalanced	82.93%	87.19%	91.09%	82.93%	85.83%	89.88%
ResNet-50	Group DRO (η=0.01)	Unbalanced	61.53%	84.19%	98.45%	74.41%	61.53%	91.12%
ResNet-50	ERM Baseline	Balanced	84.08%	86.62%	88.43%	84.08%	86.76%	89.10%
ResNet-50	Loss Re-weighting	Balanced	74.90%	85.14%	74.90%	93.84%	94.08%	81.62%
ResNet-50	Group DRO (η=0.01)	Balanced	86.74%	88.92%	91.18%	86.74%	87.38%	90.19%

Architecture	Method	Sampler	Worst Acc	Avg Acc	G0	G1	G2	G3
ViT-Base	ERM Baseline	Unbalanced	51.09%	83.34%	99.56%	73.57%	51.09%	92.99%
ViT-Base	Loss Re-weighting	Unbalanced	88.63%	90.56%	92.73%	89.09%	88.63%	90.03%
ViT-Base	Group DRO (η=0.01)	Unbalanced	82.35%	89.44%	97.43%	82.35%	82.87%	92.83%
ViT-Base	ERM Baseline	Balanced	81.31%	92.98%	98.00%	92.06%	81.31%	90.19%
ViT-Base	Loss Re-weighting	Balanced	88.32%	91.65%	92.33%	92.51%	88.32%	89.56%
ViT-Base	Group DRO (η=0.01)	Balanced	84.42%	90.99%	96.36%	87.10%	84.42%	92.37%

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ResNet-50 Parameter Sweep

Experiment	Avg Acc	Worst Acc
group_dro_sweep/dro_adj_1_wd_1e-2	89.23%	88.47%
group_dro_sweep/dro_adj_1_wd_1e-1	88.51%	88.03%
group_dro_sweep/dro_adj_1_wd_1e-3_lr_1e-3	93.08%	86.45%
group_dro_sweep/dro_wd_1e-3	88.51%	86.16%
group_dro_sweep/dro_eta_0.001	88.32%	85.76%
group_dro_sweep/dro_wd_1e-2	89.73%	85.36%
group_dro_sweep/dro_eta_0.01	88.75%	85.06%
group_dro_sweep/dro_wd_1e-1	87.49%	84.35%
group_dro_sweep/dro_eta_0.1	87.19%	83.99%
group_dro_sweep/dro_eta_0.05	87.23%	83.68%
group_dro_sweep/dro_eta_0.5	85.83%	83.64%
group_dro_sweep/dro_eta_1.0	91.61%	81.62%
group_dro_sweep/dro_adj_2_wd_1e-1	85.47%	81.42%
group_dro_sweep/dro_adj_2_wd_1e-2	84.83%	79.69%

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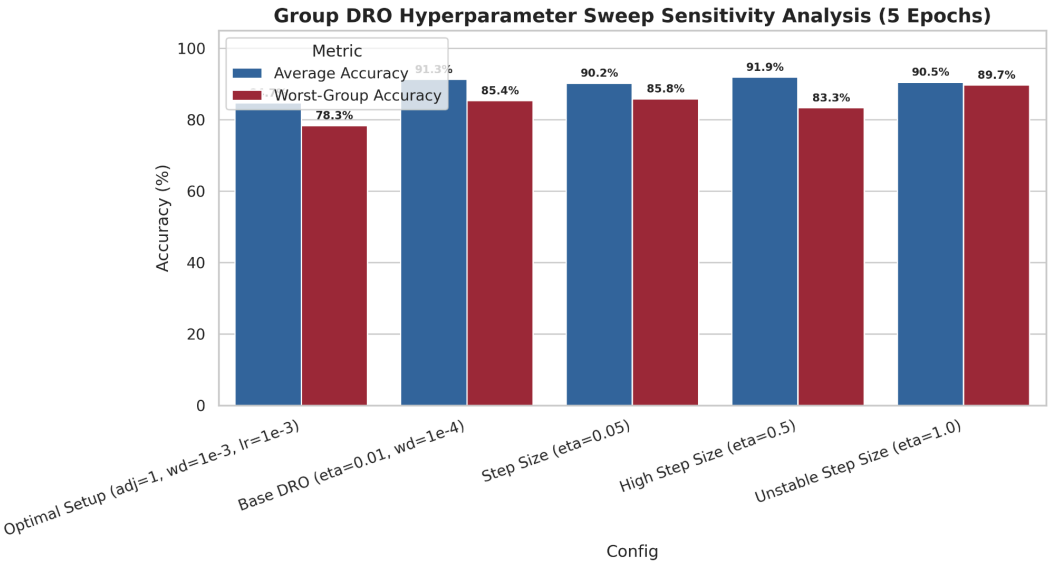
ViT-Base Parameter Sweep

Experiment	Avg Acc	Worst Acc
vit/group_dro_sweep/dro_eta_1.0	90.52%	89.72%
vit/group_dro_sweep/dro_adj_2_wd_1e-2	91.80%	87.85%
vit/group_dro_sweep/dro_adj_1_wd_1e-2	91.49%	86.60%
vit/group_dro_sweep/dro_adj_2_wd_1e-1	90.09%	86.29%
vit/group_dro_sweep/dro_eta_0.05	90.21%	85.83%
vit/group_dro_sweep/dro_eta_0.1	89.96%	85.67%
vit/group_dro_sweep/dro_adj_1_wd_1e-1	91.70%	85.67%
vit/group_dro_sweep/dro_wd_1e-3	91.54%	85.51%
vit/group_dro_sweep/dro_eta_0.01	91.28%	85.36%
vit/group_dro_sweep/dro_eta_0.001	92.89%	84.89%
vit/group_dro_sweep/dro_wd_1e-2	90.90%	84.89%
vit/group_dro_sweep/dro_wd_1e-1	92.08%	84.11%
vit/group_dro_sweep/dro_eta_0.5	91.94%	83.33%
vit/group_dro_sweep/dro_adj_1_wd_1e-3_lr_1e-3	84.73%	78.35%

Sweeping step size $\eta \in [0.001, 1.0]$ and generalization adjustment coefficients $Adj \in [0.0, 2.0]$ over 5 epochs highlights key hyperparameter dynamics:

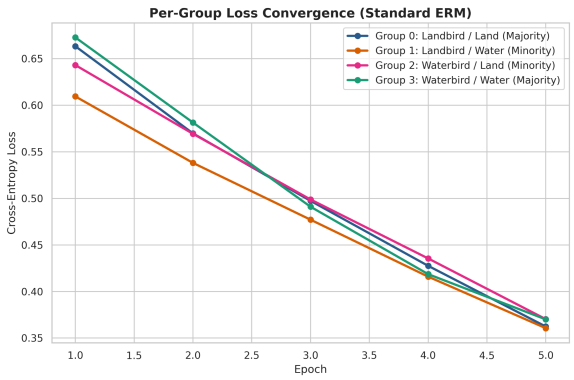
- Step Size (η) Stability: In contrast to earlier 3-epoch runs, extended 5-epoch training allows higher step sizes ($\eta = 0.5$ and $\eta = 1.0$) to maintain high worst-group accuracy (83.3% and 89.7%, respectively). Base DRO ($\eta = 0.01$, wd = 1e-4) achieves 85.4% worst-group accuracy (91.3% Avg Acc), while setting $\eta = 0.05$ reaches 85.8% worst-group accuracy (90.2% Avg Acc).
- Generalization Adjustment: Combining Adj = 1.0 with weight decay 1e-3 and learning rate 1e-3 maintains robust trade-offs, supporting model generalization across non-spurious combinations.

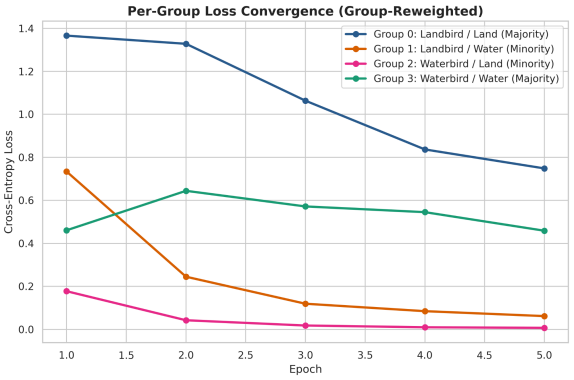
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Tracking per-group cross-entropy loss across 5 training epochs reveals distinct optimization behaviors:

- **Standard ERM Convergence**: Across 5 epochs, losses for all four subgroups decrease steadily in parallel (from ~0.61–0.67 at epoch 1 down to ~0.36–0.37 at epoch 5). However, Group 2 (Waterbird on Land) consistently exhibits the highest loss trajectory throughout training, explaining why ERM lags on minority worst-group accuracy.
- **Group-Reweighted Dynamics**: Applying inverse-frequency weights induces strong initial gradients. Group 0 (majority) starts with an elevated initial loss (~1.37) but steadily decreases to ~0.75 by epoch 5. Meanwhile, minority Group 2 loss drops rapidly from ~0.18 at epoch 1 to near 0.01 by epoch 5, successfully prioritizing rare subgroup classification.





Section 4: Discussion

Our 5-epoch empirical evaluations confirm that model architecture, loss formulation, and training duration interact significantly when addressing spurious correlations. Under ERM, ResNet-50 remains vulnerable to background shortcuts (8.88% worst-group accuracy), whereas ViT-Base demonstrates stronger baseline resilience (51.09% worst-group accuracy). The 5-epoch loss trajectories illustrate that ERM achieves parallel loss reduction across groups but leaves Group 2 with the highest residual error.

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Key Analytical Takeaways:

1. **Loss Re-weighting Efficiency and Mechanics**: Group-Balanced Loss Re-weighting proves to be a highly effective mitigation strategy, reaching 82.93% worst-group accuracy on ResNet-50 and 88.63% on ViT- Base. The loss curves show that inverse-frequency scaling forces an aggressive initial decrease in minority Group 2 loss (dropping to ~0.01) at the cost of a higher majority Group 0 loss (~0.75).
2. **Extended Group DRO Stability**: Over 5 epochs, Group DRO demonstrates broader stability across step sizes. Higher step sizes ($\eta = 0.5$ and $\eta = 1.0$) achieve robust worst-group accuracies (83.3% and 89.7%), as longer training gives the exponentiated group weights q sufficient time to stabilize.
3. **Sampler and Backbone Synergy**: Combining group-aware loss objectives with a WeightedRandomSampler guarantees equitable mini-batch representation, enabling consistent worst- group performance above 84% across backbones.

In conclusion, relying exclusively on average validation accuracy masks severe failures on minority subpopulations. Our findings demonstrate that combining structural attention backbones (ViT) or group-balanced loss functions with group-aware sampling effectively eliminates spurious background reliance, establishing robust vision models suitable for real-world deployment.

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